

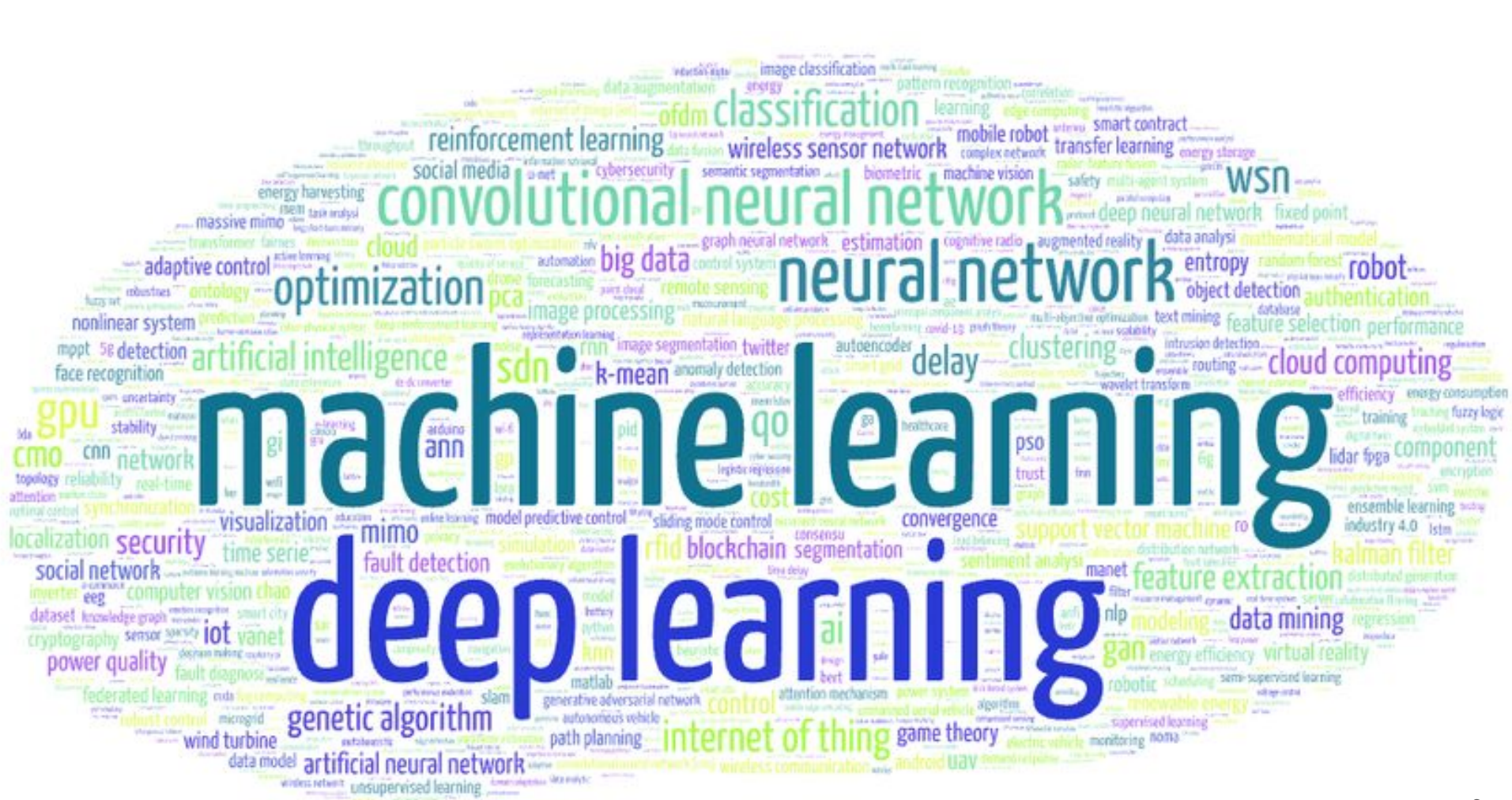
Machine Learning

Code/Astro 2026

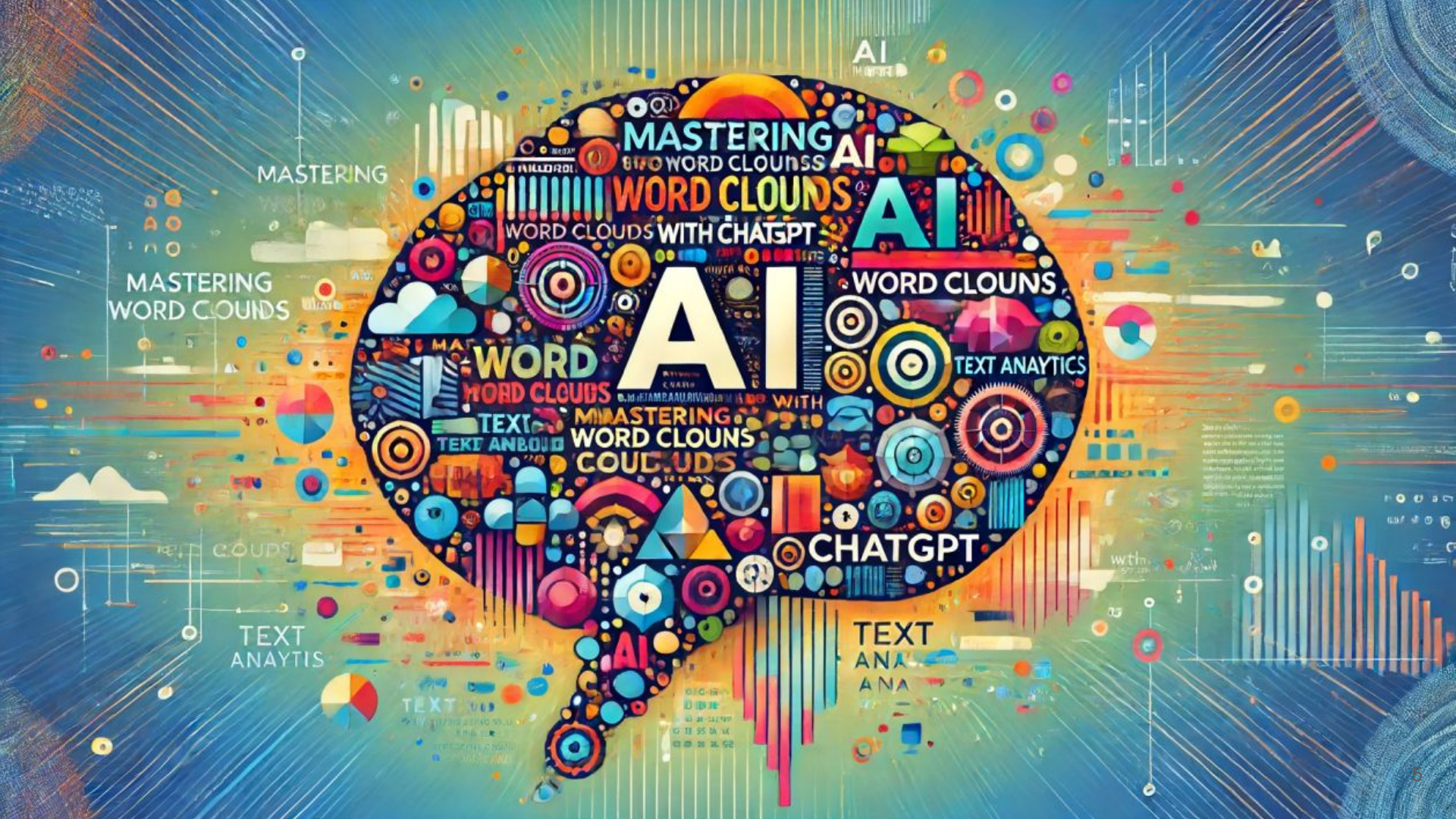


**Shh... Quiet please
machines are
learning**

What IS “Machine Learning”?



What WON'T we be covering?



AI

MASTERING

MASTERING
WORD CLOUDS

MASTERING

WORD CLOUDS

AI

WORD CLOUDS WITH CHATGPT

AI

WORD CLOUDS

WORD

WORD CLOUDS

TEXT

TEXT ANALYTICS

MASTERING

WORD CLOUDS

COULD

CHATGPT

TEXT
ANALYTICS

TEXT
ANA
ANA

Goals

Give an outline of the basics of how Machine Learning works

... empower you to learn more

Introduce some applications in astronomy research

... help you think about applications to your research

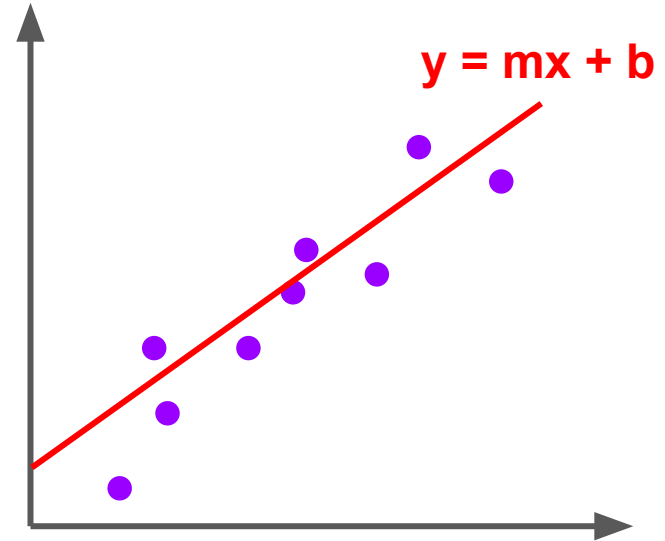
Basic Concepts

Linear Regression

Yes, Linear Regression is ML!

What?

Fit a **line** (of the form $y = mx + b$) to **data**

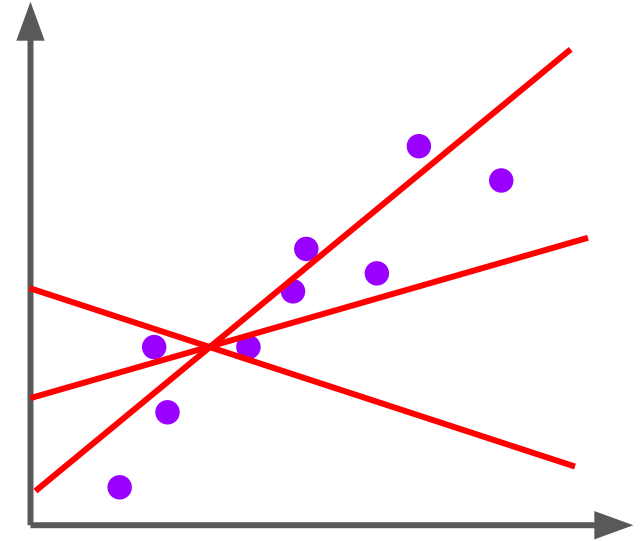


Linear Regression

Yes, Linear Regression is ML!

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Linear Regression

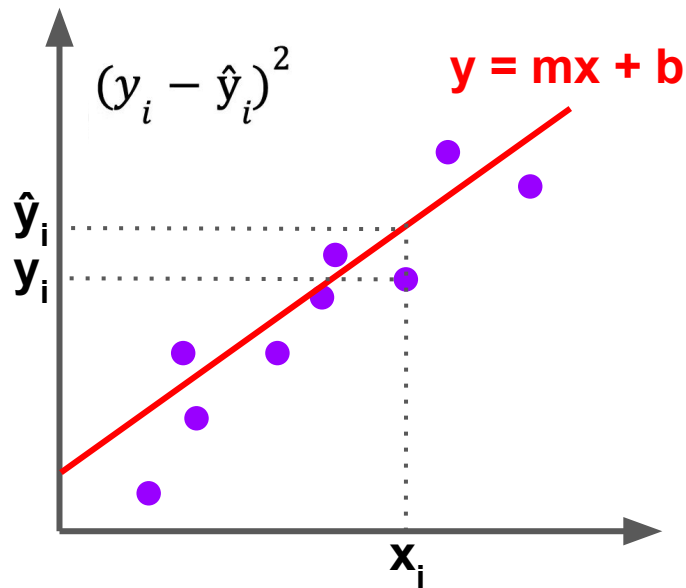
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What?

Fit a **line** (of the form $y = mx + b$) to **data**

How?

By **minimizing** a **loss function** (here, it's called the mean-squared error)



Linear Regression

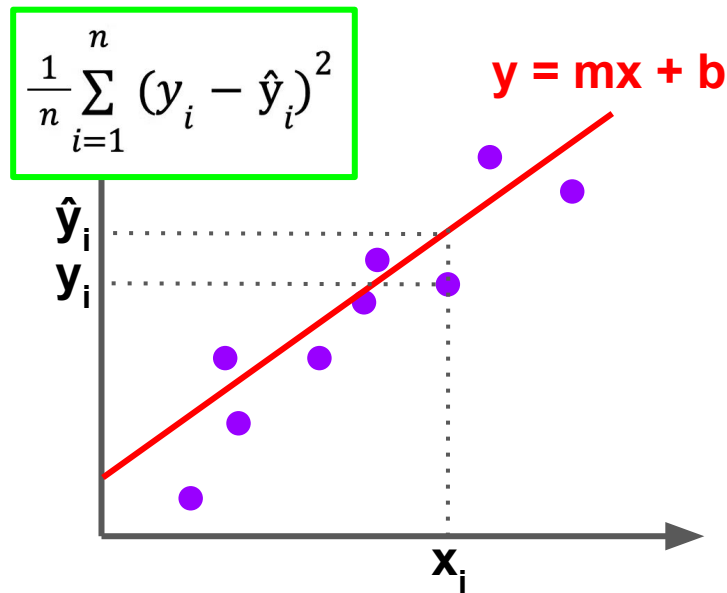
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Linear Regression

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What?

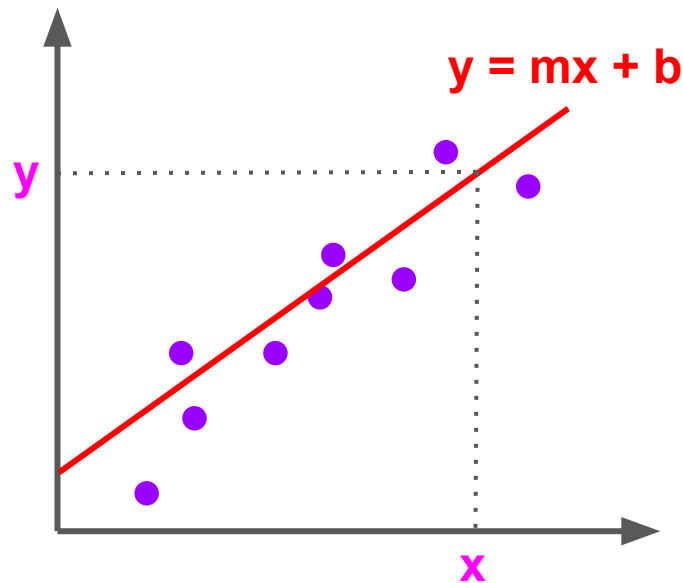
Fit a **line** (of the form $y = mx + b$) to **data**

How?

By **minimizing** a **loss function** (here, it's called the mean-squared error)

Why?

In order to **make predictions** about future data



Vocabulary

“Line” → Model with parameters m, b

“Data” → n -dimensional

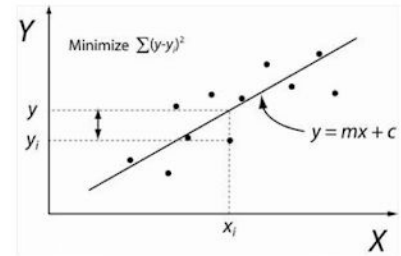
“Loss function” → Training objective

“Minimizing” → Optimization procedure

“Make predictions” → Perform some task

YOU'VE SEEN

Linear Regression

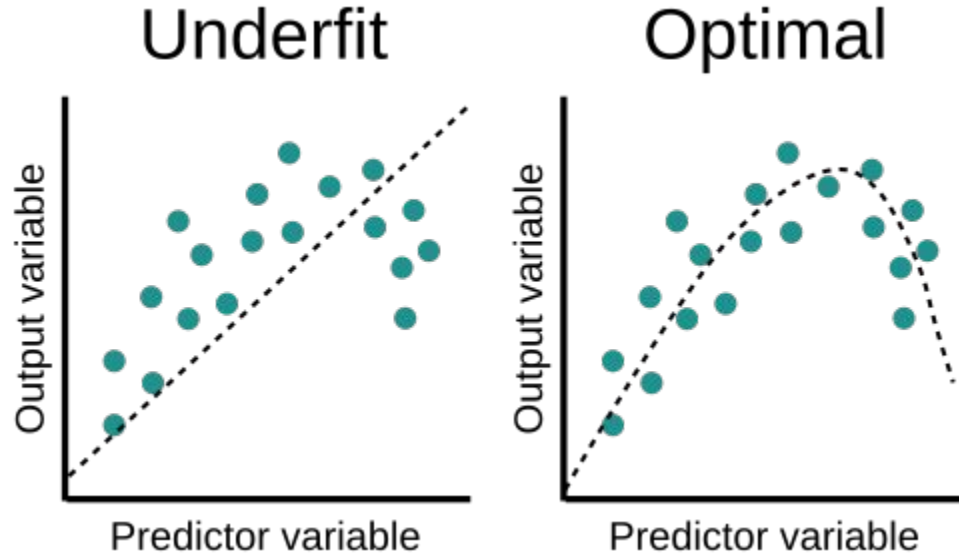


NOW GET READY FOR

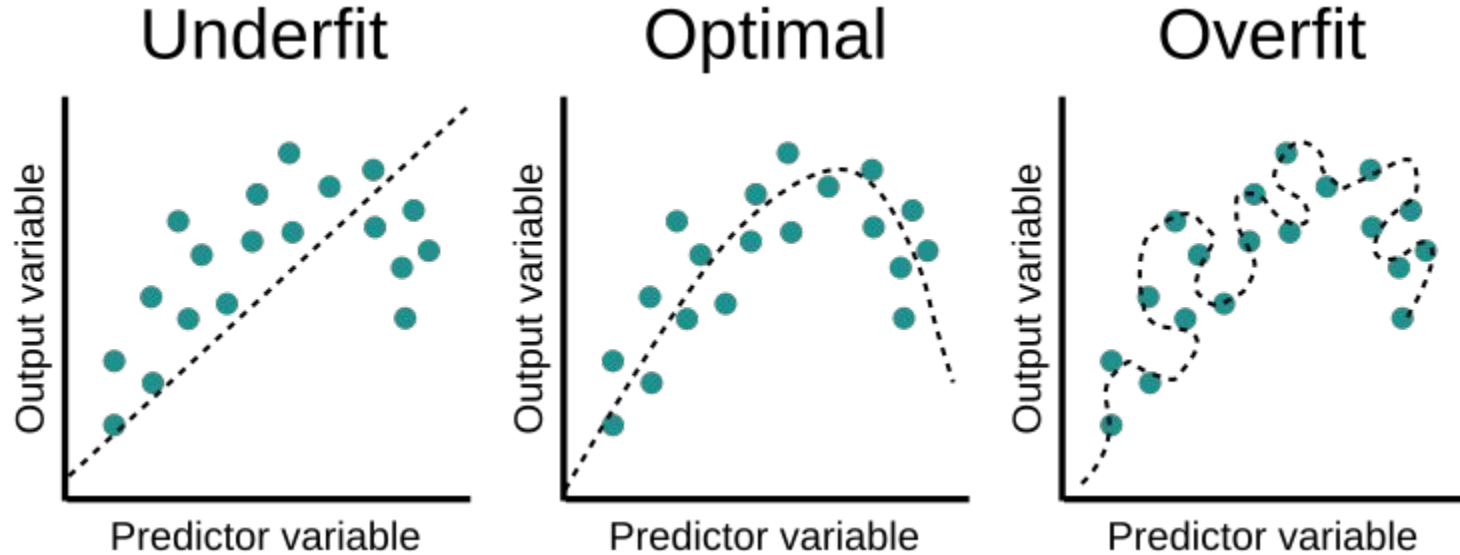
PAIN



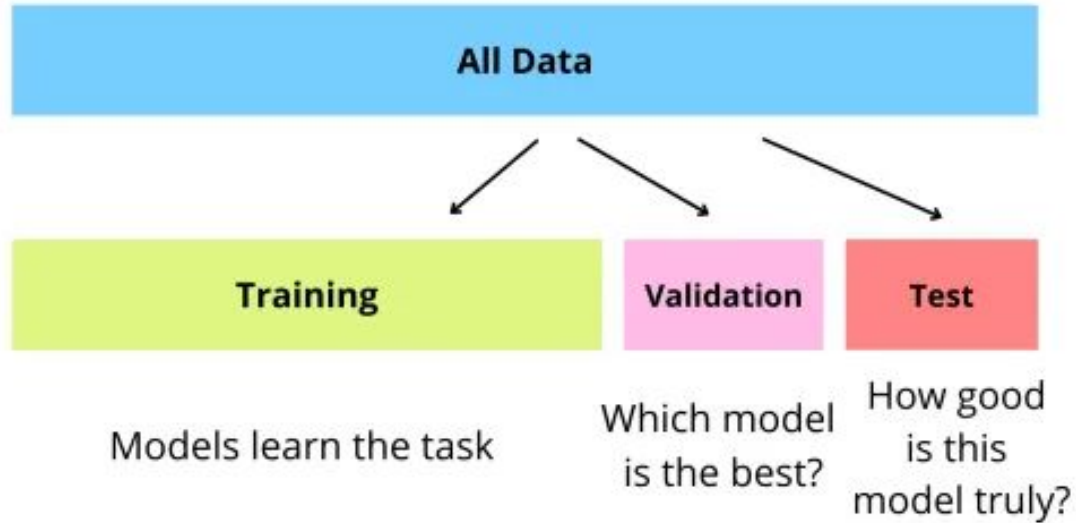
How do we choose the right **model**?



How do we choose the right **model**?



How should we split the **data**?



Machines learn by **optimizing** a **loss function**

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Learn more about loss functions here: <https://www.geeksforgeeks.org/machine-learning/ml-common-loss-functions/>

The Machine Learning Recipe:

1. Choose a **model**
2. Define a **loss function**
3. **Optimize** parameters
4. **Evaluate** on unseen **data**

Making predictions

Regression Examples

- Income vs. happiness
- Caffeine intake vs. productivity
- Hubble's Law: galaxy distance vs. recessional velocity



Making predictions

Regression Examples

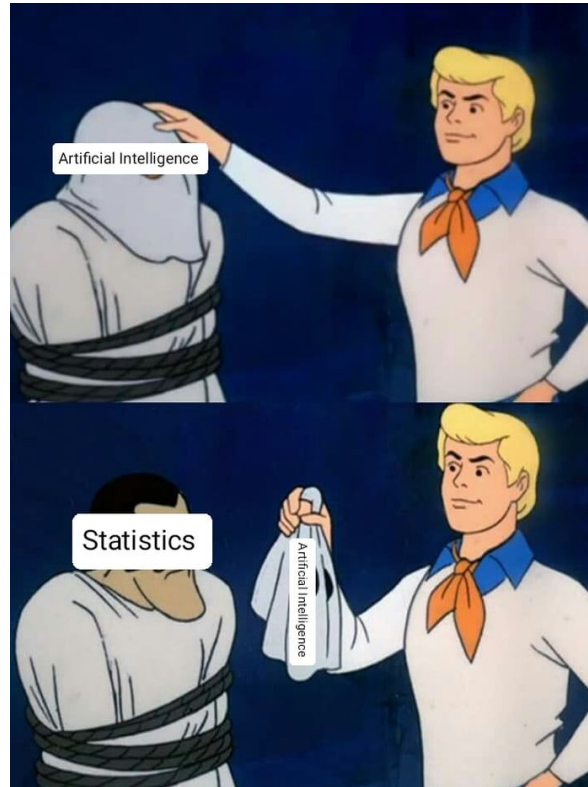
- Income vs. happiness
- Caffeine intake vs. productivity
- Hubble's Law: galaxy distance vs. recessional velocity

Classification Examples

- Identification of digit or character in a hand-drawn image
- Classifying galaxies by morphology
 - *Assuming we have some group of galaxies with known classes*
- Determining habitability of planets from period, distance, and mass

“Supervised” methods

Wait, but isn't this just statistics?



Supervised Methods

The labels or correct answer is known
(at least for the training data)

Unsupervised Methods

More “abstract” tasks often
related to uncovering patterns
or structure

Unsupervised Methods

Anomaly Detection

- Given a new observation, determine how weird it is

Clustering

- Given a large dataset *with no labels*, group similar data together

Data Generation

- Given a picture of a galaxy, generate the spectrum

Dimensionality Reduction

- Direct imaging: starlight vs. exoplanet
 - PCA-KLIP

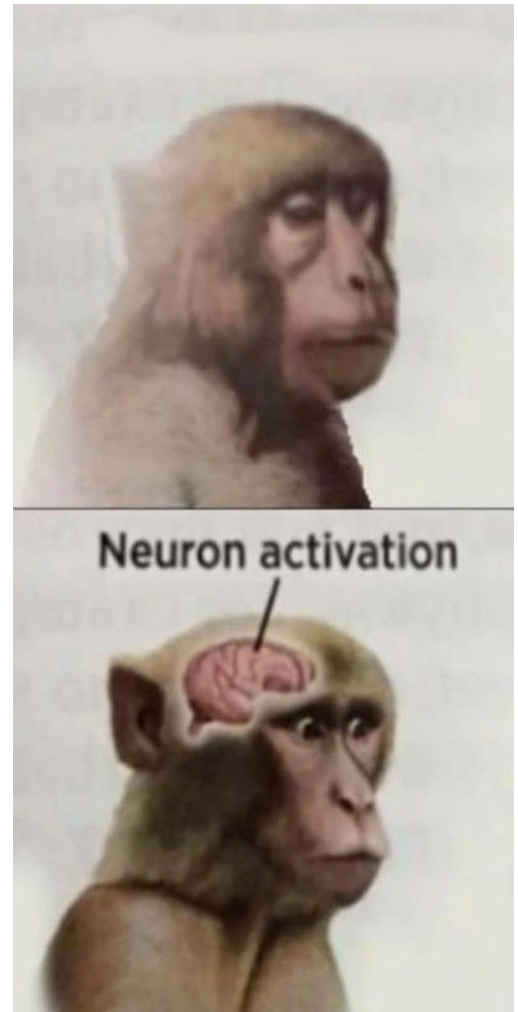
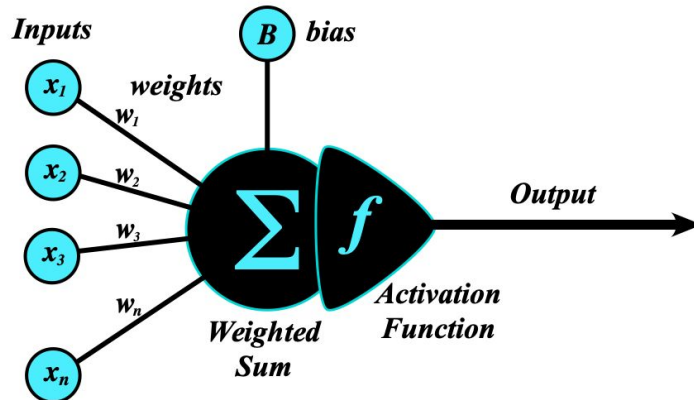
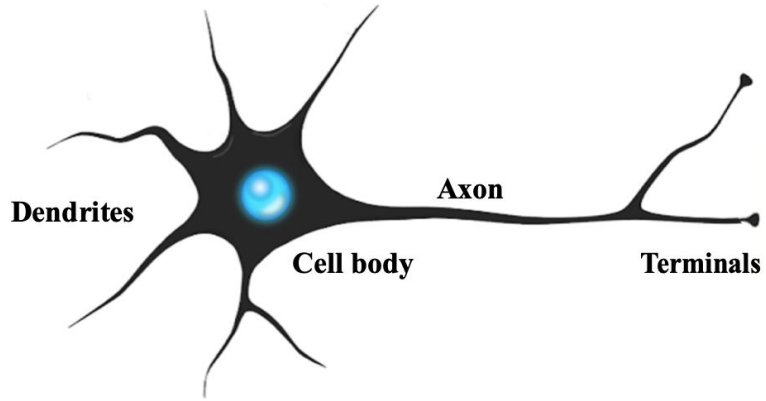
Emulation

- Replace more complicated models i.e. complicated N-body cosmological simulations

Any questions so far?

Neural Networks

Neurons



Data input

x_1

x_2

.

.

.

x_n

Data input

x_1

x_2

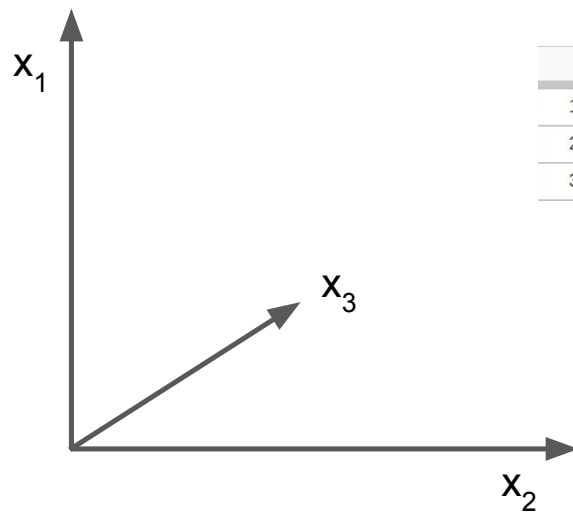
.

.

.

x_n

Many dimensions



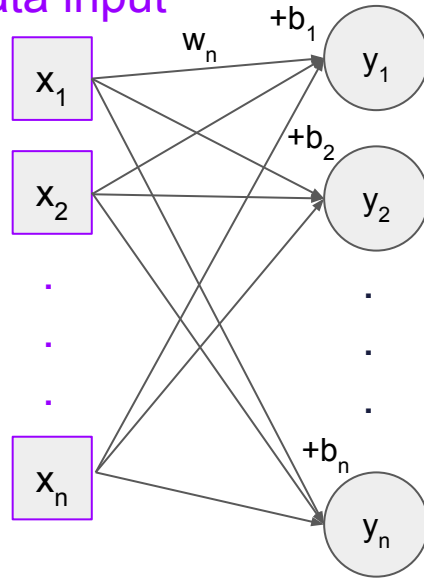
Many features

	x_1	x_2	x_3	x_4
1				
2				
3				

Hidden layer

weight + bias

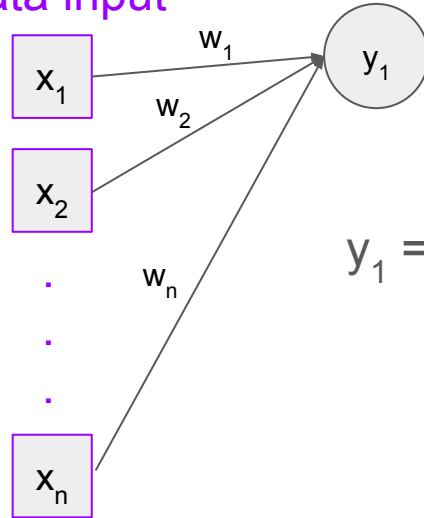
Data input



Neuron

weight + bias

Data input

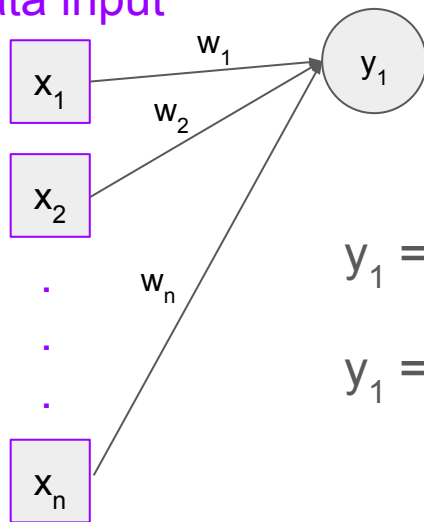


$$y_1 = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b_1$$

Neuron

weight + bias

Data input



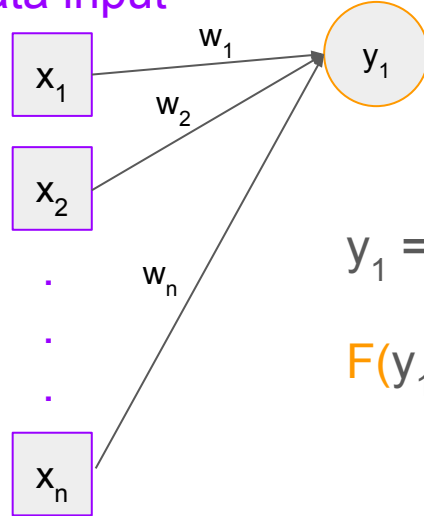
$$y_1 = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b_1$$

$$y_1 = \mathbf{w} \cdot \mathbf{x} + b_1$$

Neuron

Activation function(weight + bias)

Data input

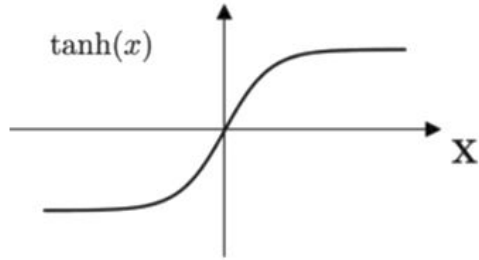


$$y_1 = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b_1$$

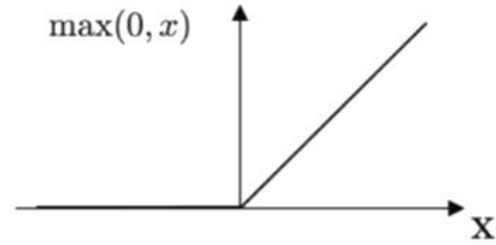
$F(y_1)$

Activation Functions

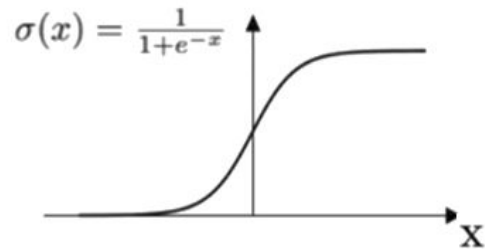
Tanh



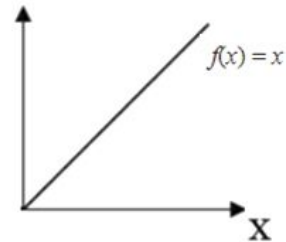
ReLU



Sigmoid

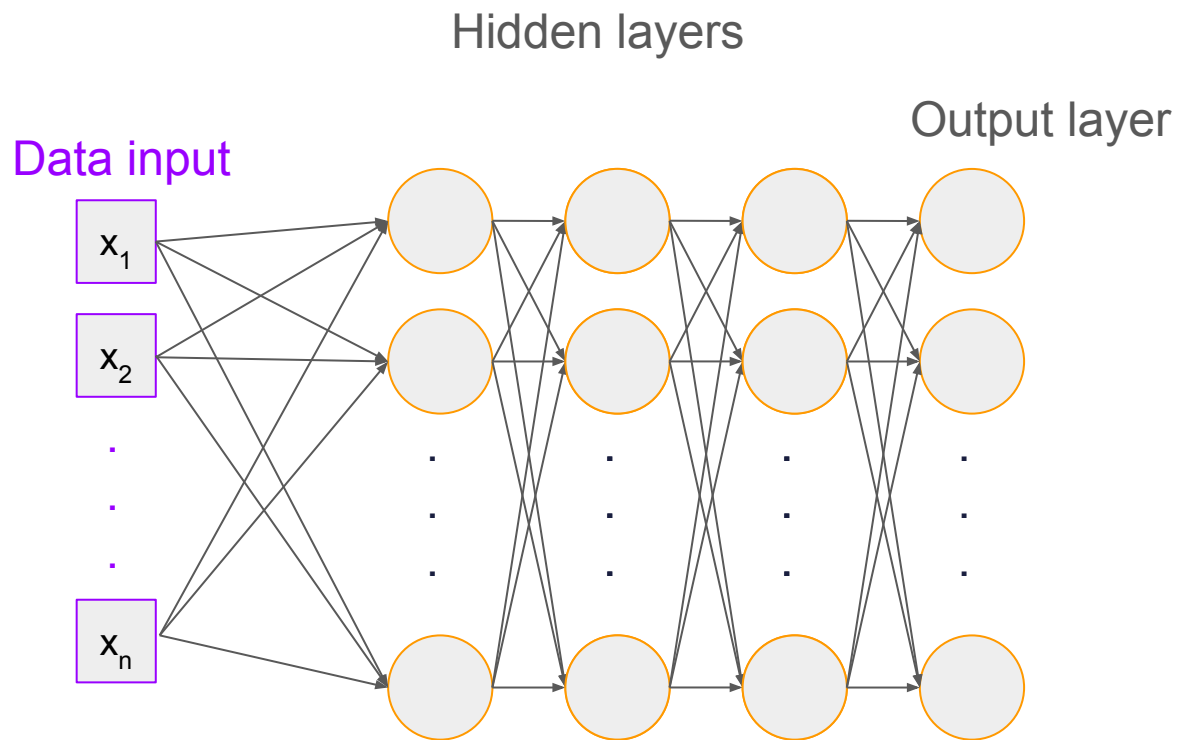


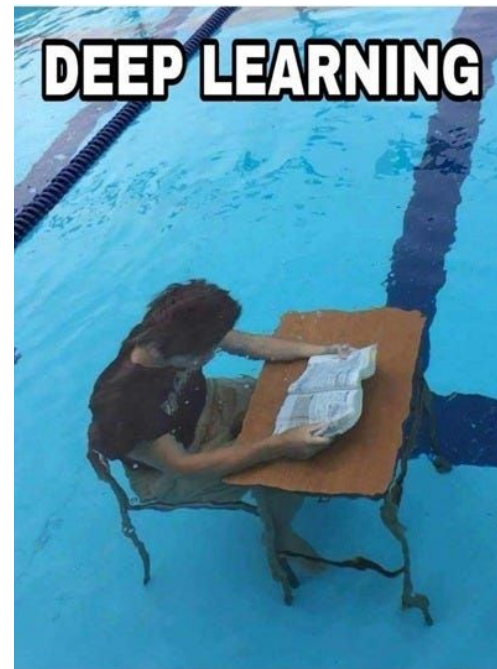
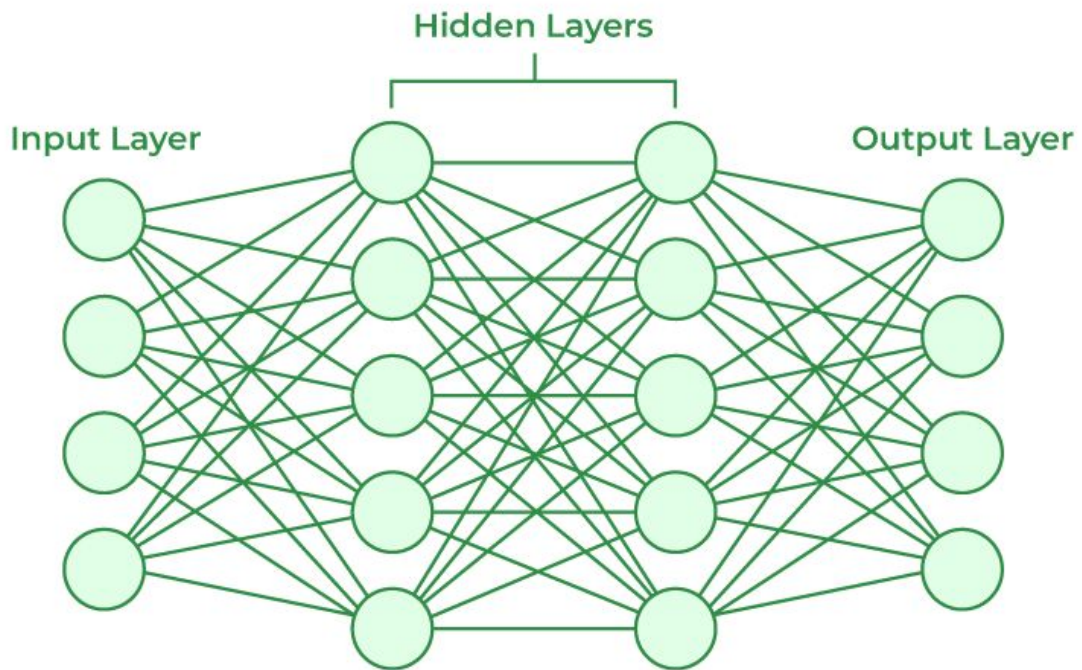
Linear



Universal Approximation Theorem

A neural network can approximate ANY continuous function, given at least *one hidden layer* and a *non-linear activation function*





The Machine Learning Recipe:

1. Choose a **model**
2. Define a **loss function**
3. **Optimize** parameters
4. **Evaluate** on unseen **data**

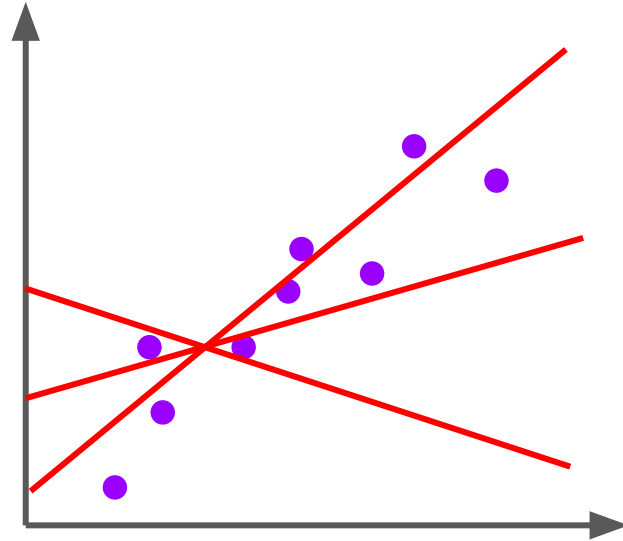
Neural network

depends on task, but
sometimes MSE

Gradient Descent

...profit B-)

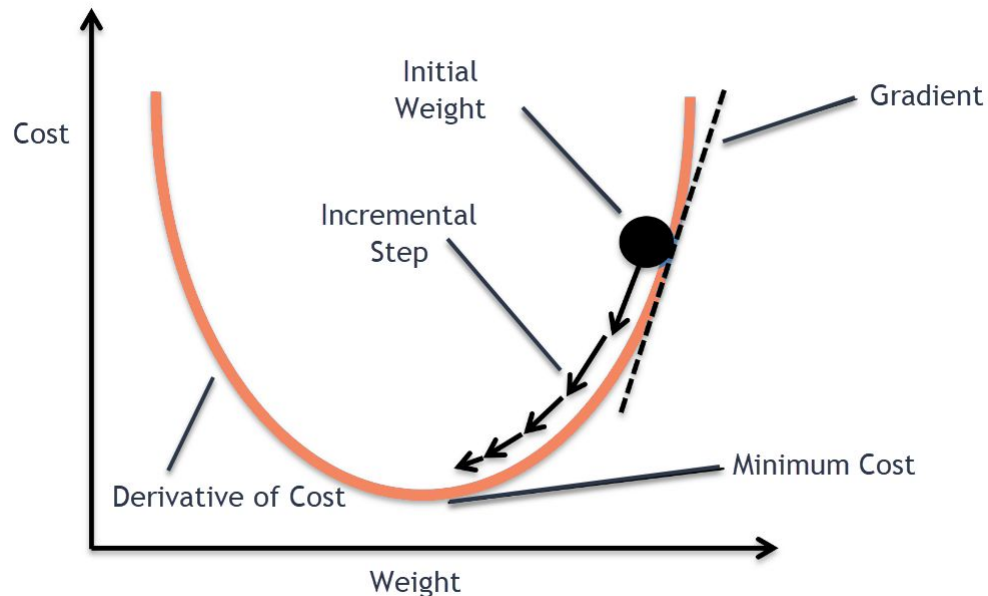
Initially, weights and biases are set to be random



Gradient Descent

How do we optimize the parameters?

1. Change them a little bit
2. Calculate how much the loss has changed
3. Was that a good step to take towards minimizing loss? Maybe go a little bit further in that direction...
 - a. But be careful to not get stuck in a local minimum
4. Repeat
 - a. The Learning Rate dictates how big a step to take



Let's learn!

LEARNING



**LEARNING
MACHINE
LEARNING**



**USING MACHINE
LEARNING TO LEARN
MACHINE LEARNING**



**LEARNING TO
USE MACHINE
LEARNING TO LEARN
MACHINE LEARNING**



Pick any from:

1. [NN Playground webtool](#)

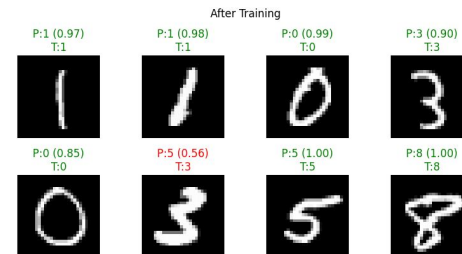
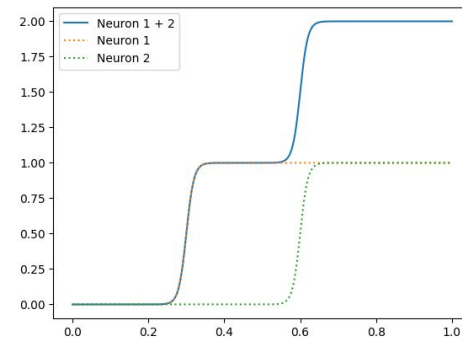
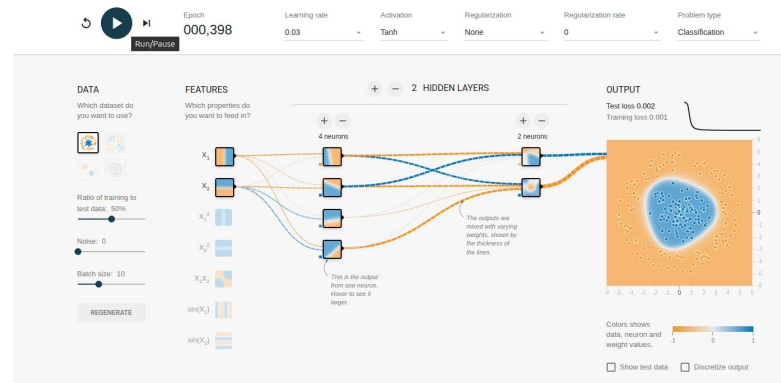
- Very fast and great visuals

2. Neurons notebook - [neurons.ipynb](#)

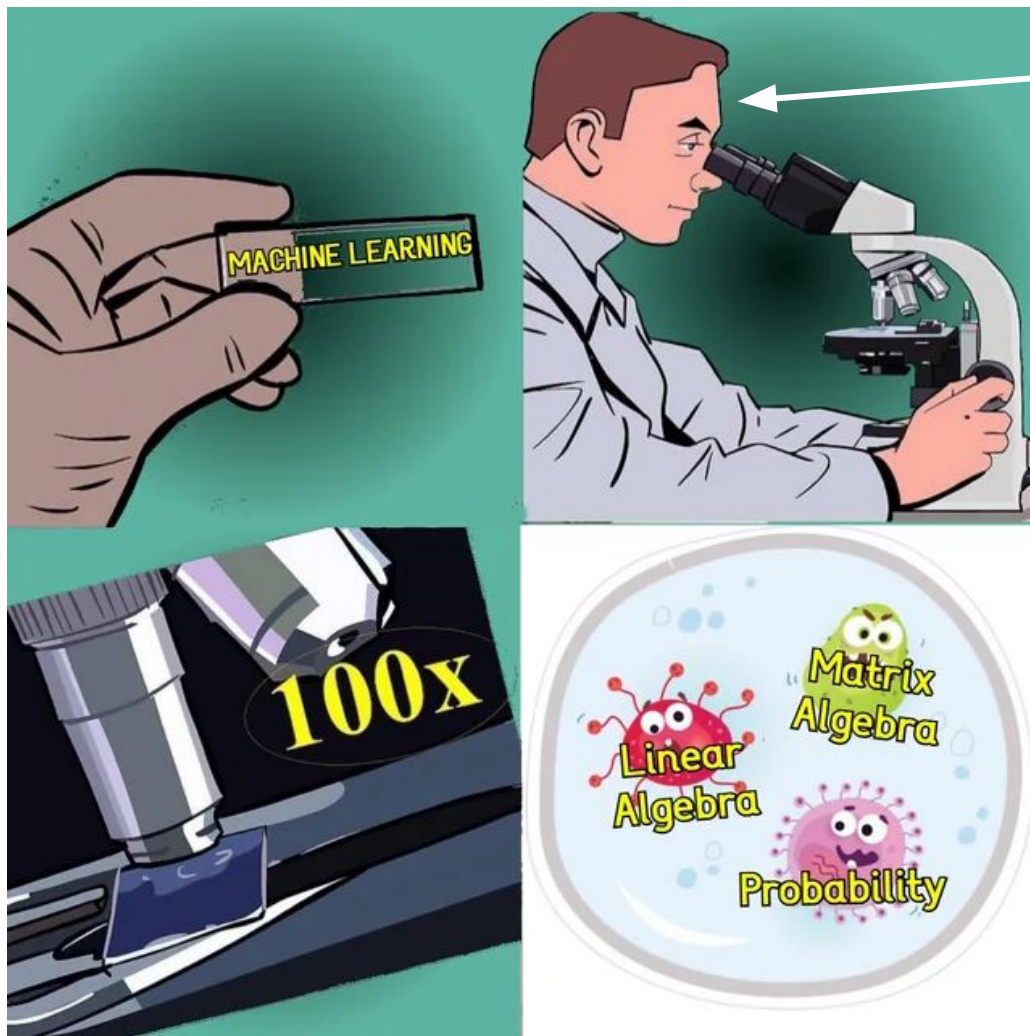
- For a deeper understanding on what neurons actually do

3. MNIST Classifier notebook - [MNIST.ipynb](#)

- If you want to build your own NN
- If it takes a while to run, go back to the other tools :)

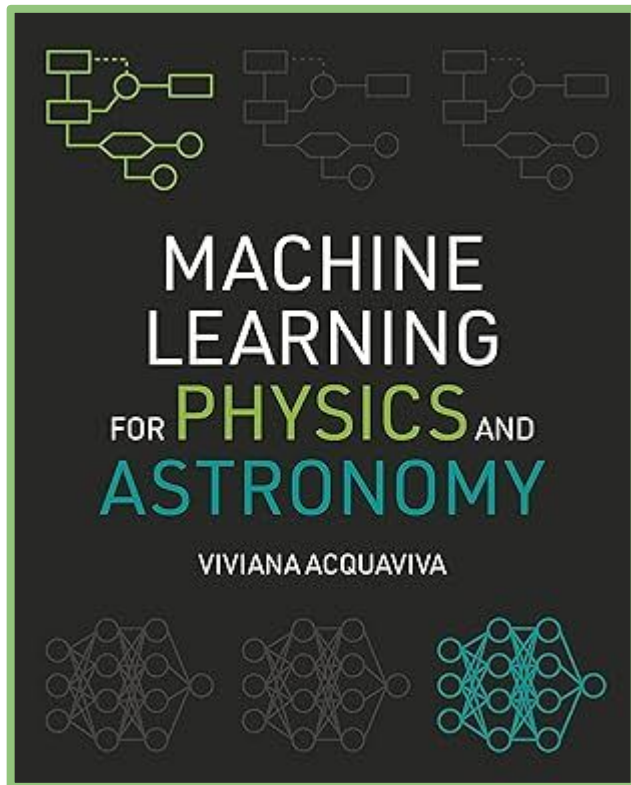






You! :)

Extra Resources!



License: CC BY-NC-SA 4.0
arXiv:2502.08222v1 [astro-ph.IM] 12 Feb 2025

How to set up your first machine learning project in astronomy

Johannes Buchner

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Sotiria Fotopoulou

H.H. Wills Physics Laboratory, University of Bristol, Tyndall
Avenue, Bristol, BS8 1TL, United Kingdom

Deep Learning

deeplearningbook.org

An MIT Press book

Ian Goodfellow and Yoshua Bengio and Aaron Courville

tingyuansen.github.io/NASA_AI_ML_STIG

NASA COSMIC ORIGINS PROGRAM

AI/ML Science & Technology Interest Group